

INDUSTRIAL PUMP P-102

Maintenance History & Service Records

Machine ID: P-102 | Equipment: Centrifugal Industrial Pump

1. Equipment Profile

P-102 is a centrifugal process pump used for continuous industrial fluid transfer. Routine condition monitoring focuses on the mechanical seal, bearings, coupling, lubrication, casing, piping connections, and mounting.

2. Historical Maintenance Records

Date	Component	Observation / Work	Status
2026-05-10	Bearings	Routine inspection completed; no visible damage reported.	Closed
2026-06-15	Mechanical seal	Minor leakage reported near seal area. Seal inspected and service completed.	Completed
2026-07-08	Coupling	Alignment checked during scheduled maintenance.	Closed
2026-08-12	Lubrication	Lubrication level checked and service performed according to maintenance schedule.	Closed

3. Recurring Concern

The June 2026 maintenance record is relevant when investigating a new visual observation of possible leakage around the mechanical seal. A new observation should be compared with the previous seal service record rather than treated as proof of a recurring failure.

4. Current Inspection Priorities

If visual analysis indicates possible leakage around the seal area, prioritize: mechanical seal condition, seal surroundings, shaft condition, lubrication, bearing condition, and coupling condition. Record whether the condition is new, unchanged, or worsening.

5. Maintenance Decision Support

A possible seal-related abnormality with no other warning signs may justify a scheduled inspection. Significant leakage, worsening condition, abnormal vibration/noise, overheating, or safety concerns should increase priority and require qualified personnel to determine the appropriate operational response.

6. Record-Keeping

New findings should be added to the maintenance history with the machine ID, date, observation, action taken, priority, and outcome. This history can be used as contextual evidence by the INNOVEXA retrieval system.

Note: These records are fictional demonstration data created for the INNOVEXA hackathon demo.